Computer Networks — Interview Questions

Networking questions asked in fresher technical interviews — models, protocols, addressing and web basics — with short answers to say aloud.

Answer like this in the interview

Define in one line → **explain** with one real-world example → add **one detail** (a difference, a use case or a limitation). Stop at 30–45 seconds and let the interviewer ask the follow-up.

Questions marked ★ **MOST ASKED** come up in almost every fresher technical round — learn those first.

Top questions to learn first

OSI model layers · OSI vs TCP/IP · TCP vs UDP · What happens when you type a URL · DNS · HTTP vs HTTPS · IP address & subnet mask · Hub vs switch vs router · TCP 3-way handshake · Common port numbers

1 · Basics

1 What is a computer network? What are its types?

Two or more devices connected to **share data and resources**.

- **PAN** — personal area (Bluetooth earbuds). • **LAN** — a building or campus.
- **MAN** — a city. • **WAN** — countries/continents; the Internet is the largest WAN.

Network topologies

Topology	Layout	Pro / con
Bus	All devices on one cable	Cheap · one cable fault stops everything
Star	All devices connect to a central switch/hub	Easy to manage · the centre is a single point of failure
Ring	Each device connects to two neighbours in a loop	Orderly · one break can stop the ring
Mesh	Every device connects to many others	Very reliable · expensive cabling
Tree / hybrid	Combination of star and bus	Scalable · complex

2 What is bandwidth, latency and throughput?

Bandwidth is the maximum data rate a link can carry (e.g. 100 Mbps). **Latency** is the delay for data to travel from source to destination (ms). **Throughput** is the actual data rate achieved, which is usually less than bandwidth.

2 · Network models

OSI model — 7 layers ★ MOST ASKED

#	Layer	Job	Protocols / devices	Data unit
7	Application	Services to the user's apps	HTTP, HTTPS, FTP, SMTP, DNS	Data
6	Presentation	Format, encryption, compression	SSL/TLS, JPEG, ASCII	Data
5	Session	Start, manage and end sessions	RPC, NetBIOS	Data
4	Transport	End-to-end delivery, ports, reliability	TCP, UDP	Segment
3	Network	Logical addressing and routing	IP, ICMP · Router	Packet
2	Data Link	MAC addressing, framing, error detection	Ethernet, ARP · Switch	Frame
1	Physical	Transmit raw bits over the medium	Cables, Wi-Fi signals · Hub	Bits

Memory aid

Top to bottom: **A**ll **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing (Application → Physical).

OSI vs TCP/IP model

Point	OSI	TCP/IP
Layers	7	4 — Application, Transport, Internet, Network Access (Link)
Nature	Reference / theoretical model	Practical model the Internet actually uses
Mapping	Application + Presentation + Session	= Application layer
	Data Link + Physical	= Network Access layer

3 · Transport layer

TCP vs UDP ★ MOST ASKED

Point	TCP	UDP
Connection	Connection-oriented (handshake first)	Connectionless
Reliability	Guaranteed, in order, retransmits lost data	No guarantee, no ordering
Speed / overhead	Slower, 20-byte header	Faster, 8-byte header
Flow & congestion control	Yes	No
Used by	Web (HTTP/HTTPS), email, file transfer, SSH	Video calls, live streaming, online games, DNS queries

3 **Explain the TCP three-way handshake.** ★ MOST ASKED

How TCP sets up a connection before sending data:

- Client → Server: **SYN** (“I want to connect”, with a sequence number).
- Server → Client: **SYN-ACK** (“OK, I acknowledge”).
- Client → Server: **ACK** — connection established.

Closing uses a four-way exchange: FIN, ACK, FIN, ACK.

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4 **What is a port number? Name common ports.** ★ MOST ASKED

A port identifies a specific **application or service** on a device; IP address + port = a **socket**. Ports range 0–65535; 0–1023 are well-known.

- HTTP **80** · HTTPS **443** · FTP **20/21** · SSH **22** · Telnet **23** · SMTP **25** · DNS **53** · DHCP **67/68** · POP3 **110** · IMAP **143** · MySQL **3306**.

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4 · Network layer & addressing

5 **What is an IP address? IPv4 vs IPv6?** ★ MOST ASKED

A logical address that identifies a device on a network.

IPv4 is 32-bit, written as four decimal numbers (192.168.1.10) — about 4.3 billion addresses. **IPv6** is 128-bit, written in hexadecimal (2001:db8::1) — practically unlimited addresses, built-in security (IPsec) and no need for NAT.

IPv4 address classes

Class	First octet	Default subnet mask	Use
A	1 – 126	255.0.0.0 (/8)	Very large networks
B	128 – 191	255.255.0.0 (/16)	Medium networks
C	192 – 223	255.255.255.0 (/24)	Small networks
D	224 – 239	—	Multicast
E	240 – 255	—	Experimental

6 What are private IP addresses, NAT and loopback?

Private ranges used inside LANs, not routable on the Internet: **10.0.0.0/8**, **172.16.0.0/12**, **192.168.0.0/16**.

NAT (Network Address Translation) lets many private devices share one public IP. **127.0.0.1** (localhost) is the loopback address — a device talking to itself.

7 What is a subnet mask? What is subnetting?

A subnet mask separates the **network part** from the **host part** of an IP address. In 192.168.1.10/24 (255.255.255.0), 192.168.1 is the network and .10 is the host.

Subnetting splits one network into smaller ones for better management and security. Usable hosts = **2^(host bits) − 2** (network and broadcast addresses excluded); /24 → 254 hosts, /26 → 62 hosts.

8 MAC address vs IP address? What is ARP?

A **MAC address** is a 48-bit **physical** address burnt into the network card (e.g. 00:1A:2B:3C:4D:5E), used inside a LAN (layer 2). An **IP address** is **logical**, can change and is used for routing across networks (layer 3). **ARP** finds the MAC address for a known IP address on the local network.

5 · Devices

Hub vs switch vs router ★ MOST ASKED

Device	OSI layer	How it forwards	Use
Hub	1 — Physical	Broadcasts to every port	Obsolete, simple LANs
Switch	2 — Data Link	Sends only to the right port using MAC addresses	Connect devices within a LAN
Router	3 — Network	Chooses the path using IP addresses	Connect different networks, e.g. home LAN to the Internet

9 What are a gateway, firewall, modem and access point?

- **Gateway** — connects networks that use different protocols; often the router acts as the default gateway.
- **Firewall** — hardware or software that allows or blocks traffic based on security rules.
- **Modem** — converts digital signals to analog and back for the ISP line.
- **Access point** — lets Wi-Fi devices join a wired network.

6 · Application layer & the web

10 What is DNS and how does it work? ★ MOST ASKED

The **Domain Name System** translates a domain name (google.com) into an **IP address** — the phone book of the Internet.

Lookup order: browser cache → OS cache → **resolver** (ISP) → **root server** → **TLD server** (.com) → **authoritative server** → IP returned and cached. DNS uses port **53**, mostly over UDP.

11 What happens when you type a URL and press Enter? ★ MOST ASKED

- The browser checks caches, then a **DNS lookup** finds the server's IP address.
- A **TCP connection** is opened (three-way handshake); for HTTPS a **TLS handshake** sets up encryption.
- The browser sends an **HTTP request** (GET /); the server processes it and returns an **HTTP response** (status code + HTML).
- The browser **renders** the page and fetches CSS, JavaScript and images with more requests.

HTTP vs HTTPS ★ MOST ASKED

Point	HTTP	HTTPS
Security	Plain text — can be read or changed in transit	Encrypted with **SSL/TLS**
Port	80	443
Certificate	Not needed	Needs an SSL/TLS certificate from a Certificate Authority
Use	Non-sensitive content	Logins, payments, every modern website

HTTP methods and status codes

Method	Purpose	Status code	Meaning
GET	Read data (in the URL, cacheable)	200 OK / 201 Created	Success
POST	Create / submit data (in the body)	301 / 302	Redirect
PUT	Replace a resource	400 / 401 / 403 / 404	Bad request / unauthorised / forbidden / not found
PATCH	Partly update a resource	500 / 503	Server error / service unavailable
DELETE	Remove a resource		

12 What is DHCP?

The **Dynamic Host Configuration Protocol** automatically gives a device an **IP address, subnet mask, default gateway and DNS server** when it joins a network. Steps: **D**iscover → **O**ffer → **R**equest → **A**cknowledge (DORA). Ports 67/68.

13 Name common application-layer protocols.

HTTP/HTTPS (web), **FTP** (file transfer), **SMTP** (send email), **POP3/IMAP** (receive email — IMAP keeps mail on the server), **DNS** (names), **DHCP** (addresses), **SSH** (secure remote login), **Telnet** (insecure remote login).

7 · Security basics

14 What is the difference between symmetric and asymmetric encryption?

Symmetric uses **one shared key** to encrypt and decrypt — fast (AES). **Asymmetric** uses a **public key** to encrypt and a **private key** to decrypt — slower but solves key sharing (RSA). HTTPS uses asymmetric encryption to exchange a key, then symmetric encryption for the data.

15 Name common network attacks.

Phishing (fake emails/sites to steal credentials), **DoS/DDoS** (flooding a server), **man-in-the-middle** (intercepting traffic), **malware/ransomware**, **SQL injection** and **spoofing**. Defences: firewalls, HTTPS, strong authentication, updates and user awareness.

16 What is a VPN?

A **Virtual Private Network** creates an **encrypted tunnel** over the public Internet, so data stays private and the user appears to be on the remote network — used for secure remote work.

Rapid-fire one-liners

Question	One-line answer
What is a protocol?	A set of rules for how devices communicate.
What is ping?	A tool that uses ICMP echo messages to test whether a host is reachable.
What is traceroute?	Shows the path (hops) packets take to a destination.
Unicast, multicast, broadcast?	One-to-one, one-to-group, one-to-all.
What is a socket?	IP address + port number — one end of a connection.
What is a proxy server?	An intermediary that forwards requests for clients (caching, filtering, privacy).
What is the Internet vs intranet?	Internet is public and global; intranet is a private network inside an organisation.
What is a cookie?	Small data a website stores in the browser to remember the user or session.
What is a default gateway?	The router a device sends traffic to when the destination is outside its network.
What is TTL in a packet?	A hop counter that stops packets looping forever; each router decreases it by one.

Common mistakes interviewers catch

- Mixing up the layers of switches (2) and routers (3).
- Saying UDP is never used for important things — DNS and video calls rely on it.
- Saying HTTPS is a different protocol from HTTP — it is HTTP over TLS.
- Forgetting −2 when counting usable hosts in a subnet.
- Saying a MAC address changes with the network — the IP changes; the MAC is tied to the hardware.